

Reexamining Busemann’s Wing Sweep Theory – A Modern Explanation of Classic Wartime Test Results

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The objective of this paper is to provide a historical reference for Simple Sweep Theory. We use modern CFD to recreate wind tunnel experiments conducted in Germany during the late 1930s and early 1940s. We examine the theory first proposed by Adolf Busemann who inspired both R.T. Jones’ Simple Sweep Theory and Küchemann’s Swept Wing Critical Pressure Theory. Our CFD studies show that the “quasi-2-D” test data found in Busemann’s landmark studies probably suffered from incipient tunnel choking at the higher tested sweep angles. We also show that pure 2-D and quasi-2-D data differ enough from each other to produce noticeably different results, particularly when applied to determine the Critical Mach Number of a candidate wing section.

I. Introduction

IN 1935, Adolf Busemann presented a paper at the Fifth Volta Congress that revolutionized the theory of swept wings at high speeds.[1] 1935 was an auspicious year in Italy as well as elsewhere in the developed world. Mussolini was becoming increasingly bellicose. Hitler had fully consolidated his dictatorial powers in Germany. The political situation between Japan and the United States grew ever more tense. At the Volta Congress, a broad spectrum of fluid dynamics experts met and mingled peacefully; ideas presented here rapidly gained pre-eminence in many nations.

As Busemann reflected in his own Memoir, “the theoretical aerodynamics of the Congress was divided ... into two separate fields, general methods to master the involved mathematical and physical relations, as opposed to the treatment of particular questions” while “both fields were further subdivided into both subsonic and supersonic speed ranges.”[2] Prandtl began with a general introduction to compressible flow. Britain’s G.I. Taylor spoke on the flow over a cone with zero incidence. Von Kármán spoke on drag formation at supersonic speeds. Then Busemann introduced the concept of a swept wing in purely supersonic flow, with a hope to “turn the wave-drag force somewhat away from the flight direction.”[2]

While Busemann’s mathematics concerned itself solely with supersonic freestream flows,[1] the community quickly observed that the same principles might apply to transonic flows where flow is only locally supersonic over some portion of the vehicle. Due to its basis in purely supersonic flow, Busemann’s Theory neglected any sort of three-dimensional or finite wing effects. Thus, Busemann’s Theory – which holds that the shock formation characteristics of a swept wing are **solely** determined by its leading-edge sweep – remains a “quasi-2-D” theory; see FIGURE 1 (overleaf).

Co-author Takahashi, during his time in industry realized that “quasi-2-D” simple swept wing theory was much less general than expected.[3] Doug McLean from Boeing wrote that before 3-D transonic CFD design tools became routinely available, design technologies were typically developed first for 2-D airfoils.[4] While both McLean and Takahashi realize that the “quantitative equivalence” of 2-D airfoil results on a swept, tapered wing, is limited, [3][4] the very widely cited Anderson texts [5][6] teach only “2-D Simple Sweep Theory.” Anderson maintains that the **sole**

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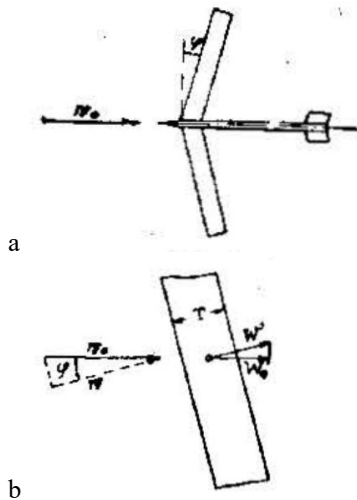


FIGURE 1 – Swept Wing Approximation from Busemann [1] a) Definition of Sweep, b) Breaking Up Incoming Flow into Components Normal and Orthogonal to the Leading Edge

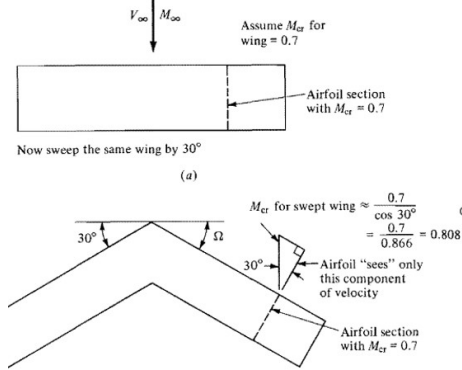


FIGURE 2 – Classic Explanation of Simple Sweep Theory from Anderson [5]

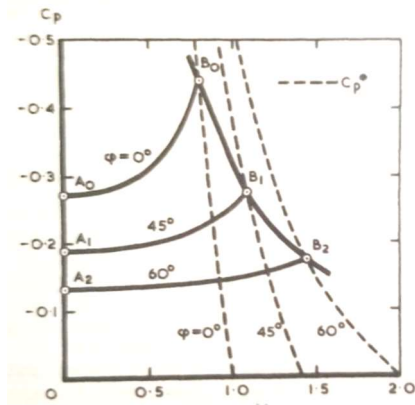


FIGURE 3 – Küchemann nuanced effects of Sweep on Critical Pressure Coefficient [11]

parameter of interest on a swept wing is the Mach number of the oncoming flow normal to the leading edge; see FIGURE 2.

In the 1950's, seminal work to develop 3-D transonic wings leveraged the concept of the “swept-2-D” airfoil.[7] This approach to design, as documented by H.H. Pearcey at the Royal Aeronautical Establishment in Farnborough, lead to the successful development of the Phase II transonic wing for the Avro Vulcan bomber.[7][8] Professor Takahashi is aware that many current wing design teams continue to base their 3-D design **directly** upon 2-D airfoil data. Widely used codes like XFLR5 adopt a hybrid 2-D superposition approach – for better or for worse – to model real aircraft with finite wings.[9][10]

Thus, seeing how important “Simple Sweep Theory” remains to many in the design community, we seek to better understand and document how “2-D Simple Sweep Theory” came to be “validated.” In a nutshell, quasi-2-D swept wing theory was a direct byproduct from Busemann’s seminal 1935 paper. Initial experimental “validation” of these ideas occurred around the time of the Second World War at newly built experimental facilities, mostly in Nazi Germany. As the Nazi regime was secretive, few reports were openly published. Fortunately, many “closely held documents” were captured by the Allies at the end of the war. In hindsight, the British consensus as found in Küchemann’s famous treatise [11] and RAE reports by and Pearcey [7], Bickley [12], Newby [13], and Haines [14], among others, reveals a more accurate, and more nuanced approach to wing sweep; see FIGURE 3

Indeed, Küchemann’s Transonic Critical Pressure Coefficient Equation, as plotted in FIGURE 3 differs from Anderson’s equation.[5] In a pair of AIAA conference papers, Kirkman & Takahashi demonstrated fundamental flaws in the commonly taught 2-D sweep theory and validated Küchemann’s equation.[15][16]

In this paper, we re-create, using modern volume-grid CFD methods, the classic German wartime test data used to validate sweep theory which is really a test of a “quasi 2-D” wing which spans the test section of a wind tunnel. We will show how these results which were once used to validate “2-D Simple Sweep Theory” upon closer inspection actually contradict it.

II. Prior Art

In this section, we critically examine the history of sweep theory and show how “Simple Sweep Theory” entered the canon of American aerodynamics education and influences many generations of aerodynamicists and designers.

A. Basic Premise of Busemann’s Simple Sweep Theory

Adolf Busemann, in his 1935 Volta Congress paper, [1] was the first to postulate that leading edge sweep could alter the speed dependent characteristics of a wing. A close reading of his writings reveals that he mentions neither subsonic nor transonic flows despite its

inclusion of figures representing an aircraft with a high aspect ratio wing; FIGURE 1. Indeed, his mathematics concerns itself solely with fully supersonic flows. Beyond the transformation of freestream Mach number into the Mach number normal to the leading edge, all of the other equations contain the term $\sqrt{M^2 - 1}$; if applied to a problem featuring subsonic freestream or subsonic-leading-edge flow this would result in an imaginary number for pressure being estimated from an unswept basis; see FIGURE 4.

B. German Applications of Busemann's Ideas to Transonic Aircraft Design

It seems lost to time the exact details regarding how Busemann's supersonic insight became applied to subsonic wings. It seems only probable that these ideas formed in a collegial environment as both Busemann and Prandtl were faculty at Göttingen. George Schairer, late of Boeing, recalled a story where Busemann went to dinner at the 1935 Volta Congress with "von Karman, Dryden, Lockspeiser, Crocco and a few others. At dinner Crocco had sketched an airplane with swept wings ... [and] labelled it "The Airplane of the Future" and passed it around the table." [17]

In any event, early German swept wing tests confirmed the trend that leading edge sweep improved the transonic characteristics of practical wings. This work was performed at the LFA (Luftfahrtforschungsanstalt) facility, the DVL (Deutsche Versuchsanstalt für Luftfahrt) facility in Berlin and at TH Braunschweig (THBS) by the young Hermann Schlichting; see FIGURE 5.[18][19] Adolf Betz and Hubert Ludweig were actively testing swept-wing models at the AVA (Aerodynamische Versuchsanstalt) wind tunnel at Göttingen.[18][20]

One important document may be found as published by the Smithsonian Institution in their series *Captured German Aeronautical Documents*. [21] This is a translation of a 1943 Lilienthal Gesellschaft report where Busemann presented an overall briefing on the state of swept wing research across multiple German research facilities. In this report he both qualitatively confirmed the benefits of leading edge sweep (φ in these figures) as a design feature which would reduce the transonic drag rise of finite wings; see FIGURE 6 and quantitatively examined the effects of sweep, freestream Mach number and normal-to-the-

In erster Näherung kann man diese Abhängigkeit auch für den ganzen Druckunterschied Δp bei nicht zu grossen Winkeln β verwenden :

$$[4] \quad \Delta p = \pm q \cdot C_1 \cdot \beta = q \cdot \frac{2\beta}{\sqrt{M^2 - 1}}$$

FIGURE 4 – Pressure equations from Busemann's 1935 paper [1]

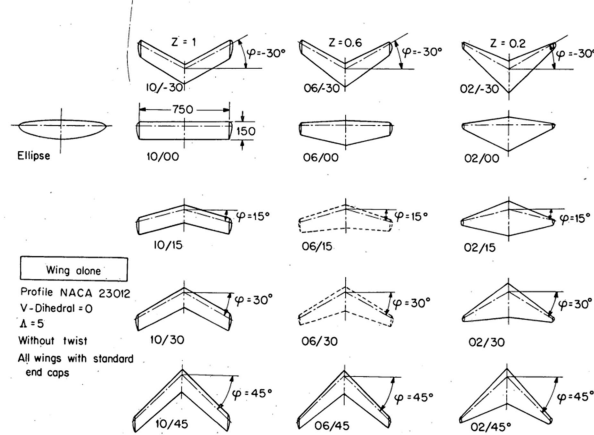


FIGURE 5 – Parametric Study of Finite Swept Wings at Braunschweig; after Schlichting [19]

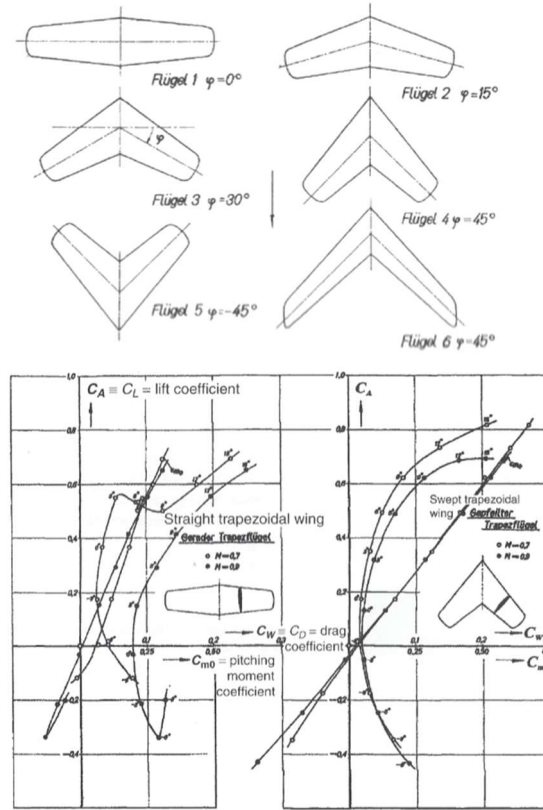


FIGURE 6 – Qualitative Confirmation that Leading Edge Sweep reduces Transonic Drag of a Wing [16]

leading-edge Mach number upon the pressure distribution of wings.[21]

Ludweig ran a comprehensive test program examining wings with various sweep angles, including one with forward (negative) sweep; see FIGURE 6.[21][22][23] Since Busemann's Theory postulated that the Mach number normal to the leading edge was the driving factor, there should be no difference between a forward swept wing or a swept back wing, in terms of delaying the drag divergence. Indeed, Ludweig found that the effects of sweep postulated by Busemann were **qualitatively** correct; he also found that the forward swept wing had almost the same aerodynamic qualities as the swept back wing. However, the reader should understand that these findings only prove that Busemann was qualitatively correct; they do not provide the full story.

We find Busemann's Lilienthal Gesellschaft report interesting because, in hindsight, we see that he realized that "for the theoretical considerations the yawed wing of infinite span was taken as a basis for the middle section of [the finite] wing." [21] He added that "it could be clearly seen ... that the results of the measurements did not longer apply;" in particular "the plane of symmetry in the center of the swept back wing produced especially pronounced deviations from the plain yawed wing of infinite span." [21]

FIGURE 7 reproduces the second set of data presented at the Lilienthal Gesellschaft.[21] Busemann realized that it was important to understand if one was truly "justified in splitting up the [flow] into an unimportant tangential component and an effective normal component." [21] By mounting a test-section spanning airfoil into the tunnel, the Braunschweig team was able to take rough center-line pressure data on both upper and lower wing surfaces at a variety of speeds and sweep angles (β in these figures). When the freestream was adjusted to maintain a constant Mach number normal to the leading edge, the "general pattern of the pressure distribution [s were] similar." [21] Thus, in this work Busemann himself admitted that the agreement between theory and test was **inexact**.

C. Basic Premise of Simple Sweep Theory from R.T. Jones

The very influential American aerodynamicist R.T. Jones also played a significant role in the development of Sweep Theory. In *Wing Theory*, he states that "Busemann's first theory was incomplete in the sense that he considered only 'supersonic' sweep." [24] He states that the realization that sweep could influence transonic performance came during wartime Germany by Busemann and Betz, and "in late 1944 by me in the United States." [24]

Schairer wrote that "late in 1944 or early 1945 [Jones] hit upon the concept of sweepback." [17] Jones "wrote a paper on the subject and submitted it to NACA for publication; it was turned down by the NACA editorial committee." He also maintains that "Jones separately invented sweepback and first brought it to the attention of U.S. Aerodynamicists." [17] Schairer gives him "full credit for the XB-47 being built with swept wings." [17]

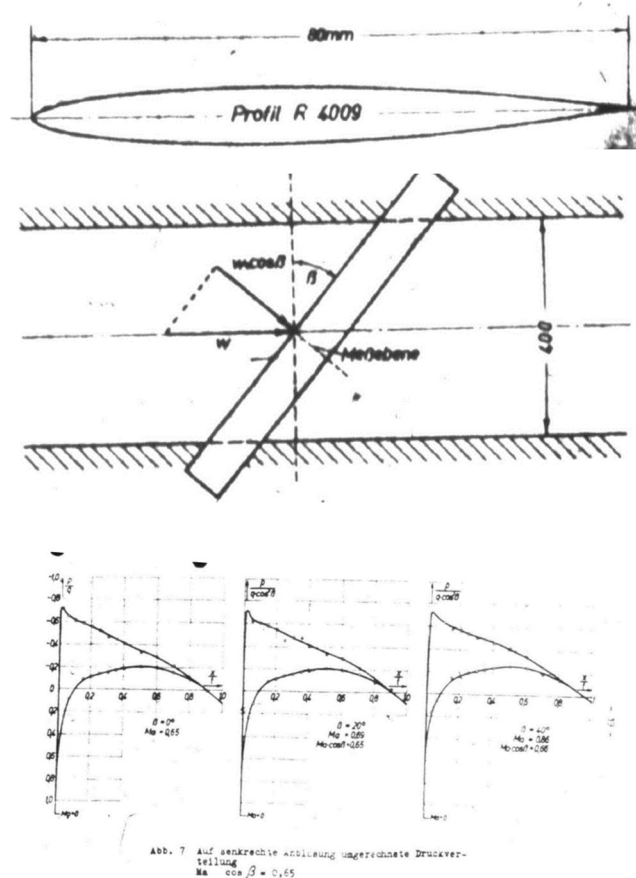


FIGURE 7 – Braunschweig Wind Tunnel Test Model Configuration from Busemann [21] and Semi-Infinite Yawed Wing Centerline pressure coefficient data

Jones specifically states that the “forces on a sufficiently long, narrow wing ... are independent of that component of the flight velocity in the direction of the long axis.”[24] He maintains that “forces then depend **only** on the reduced component velocity perpendicular to the long axis.”[24] Jones notes that Busemann did not operate in a vacuum; that Max Munk used this concept in a 1924 work on the lateral stability of straight and swept wings.[25].

R.T. Jones boils the effects of sweepback into his theory by considering that a swept finite wing may be reduced to an “approximately cylindrical airfoil” operating at sideslip.[26] From that vantage point, flow equations may be reduced assuming small disturbances across constant cross sections. He holds that if the flow patterns in “planes perpendicular” to the long axis are similar to 2-D flow – recasting the problem solely in terms of components normal to the leading edge.

$$\left[1 - \left(\frac{V \cos \varphi}{c}\right)^2\right] \frac{d^2 \Phi}{dy^2} + \frac{d^2 \Phi}{dz^2} = 0 \quad (1)$$

In this sense, Jones’ derivation follows Busemann’s original **supersonic** concept for flows around swept wings; see FIGURE 8. If we can neglect all contributions that arise from the implied spanwise flow, the velocity over the wing that matters are scaled by the cosine of the sweep angle. This is also mentioned in NACA TN 1033 where Jones states that, at supersonic speeds, the attachment of plane waves to the airfoil at near sonic or supersonic speeds (Ackeret Theory) may be avoided and the pressure drag may be reduced using planforms in which the angle of the sweepback is greater than the Mach angle.[27] Following this, we can transform the pressure coefficient, the dynamic pressure, the lift, the lift curve slope all upon the cosine of the sweep angle:

$$\frac{\Delta p}{\frac{1}{2} \rho U^2} = \frac{f(\xi)}{\sqrt{1 - (M_\infty \cos \varphi)^2}} \quad (2)$$

$$C_L = \frac{2\pi\alpha \cos \varphi}{\sqrt{1 - (M_\infty \cos \varphi)^2}} \quad (3)$$

$$\left(\frac{\delta C_L}{\delta \alpha}\right)_{M_\infty, \varphi} = \left(\frac{\delta C_L}{\delta \alpha}\right)_{0,0} \frac{(\cos \varphi)^2}{\sqrt{1 - (M_\infty \cos \varphi)^2}} \quad (4)$$

Jones goes on to mention that the analysis indicates that for aerodynamic efficiency, wings designed for flight as supersonic speeds should be swept back at an angle greater than the Mach angle and the sweep angle should be such that the component of the velocity normal to the leading edge is less than the critical speed of the airfoil sections.[25] Jones’ need to “prove” his theory became the genesis for the reuse of the Braunschweig “quasi-2-D” wing wind tunnel test which as seen in NACA TN 1107 [28], NACA TR 863 [26] and in his famous book, *High Speed Wing Theory*. [29]. Once again, we see a test where the R4009 airfoil that fully spans the test section represents a quasi-2-D swept wing. The experiment varies the sweep angle while the test section remains unchanged. From a wind axis perspective, the chord of the wing increases with the inverse of the cosine of the sweep angle. Thus, we see that more and more of the wing (area wise) is immersed in the test section as the sweep angle is increased. From a wind axis perspective, the thickness-to-chord ratio of the wing decreases with the cosine of the sweep angle.

Jones interprets the LVA Braunschweig experimental data set by first asking us to consider the implications of the pressure distributions obtained over the oblique, yawed wing He shows us pressure data where the y coordinates represent the coefficient of pressure in the leading-edge reference frame ($U \cos \beta$ in FIGURES 7 and 8 represents the component of the freestream velocity normal to the leading-edge). To transform the test data to wind axis (what we would obtain from modern CFD methods) we would need to multiply the data by $\frac{1}{\cos^2(\beta)}$.

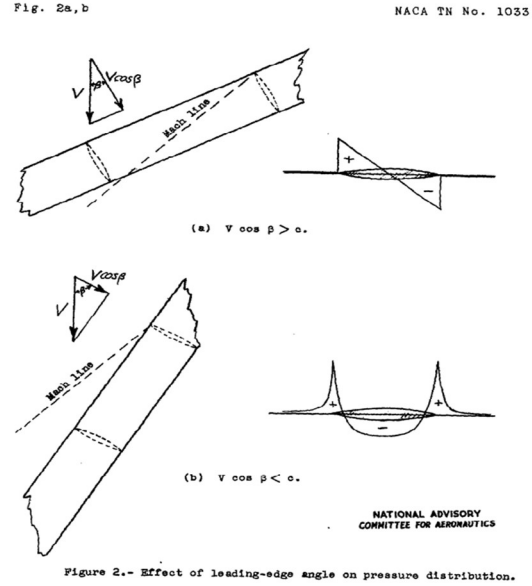


FIGURE 8 – R.T. Jones decomposition of freestream flow into components normal-to-the-leading edge and orthogonal-to-the-leading-edge.[27]

R.T. Jones' argument seems simple, if the three pressure plots shown in FIGURE 9 are essentially identical, they prove Simple Sweep Theory. He asserts that the similarity in pressure data between the three centerline pressure measurements (taken at 0° , 20° and 40° sweep) **quantitatively** confirms that 2-D Simple Sweep Theory applies – at least for this geometry. He argues that the pressure forces on the wing will act normal to its axis and therefore both the pressure drag, and the lift should be reduced by a factor proportional to the square of the cosine of the sweep angle. Finding an exact correlation where Busemann found only an approximate trend, Jones establishes the idea that Simple Sweep Theory is a universal principle which affects the flow field over swept wings.

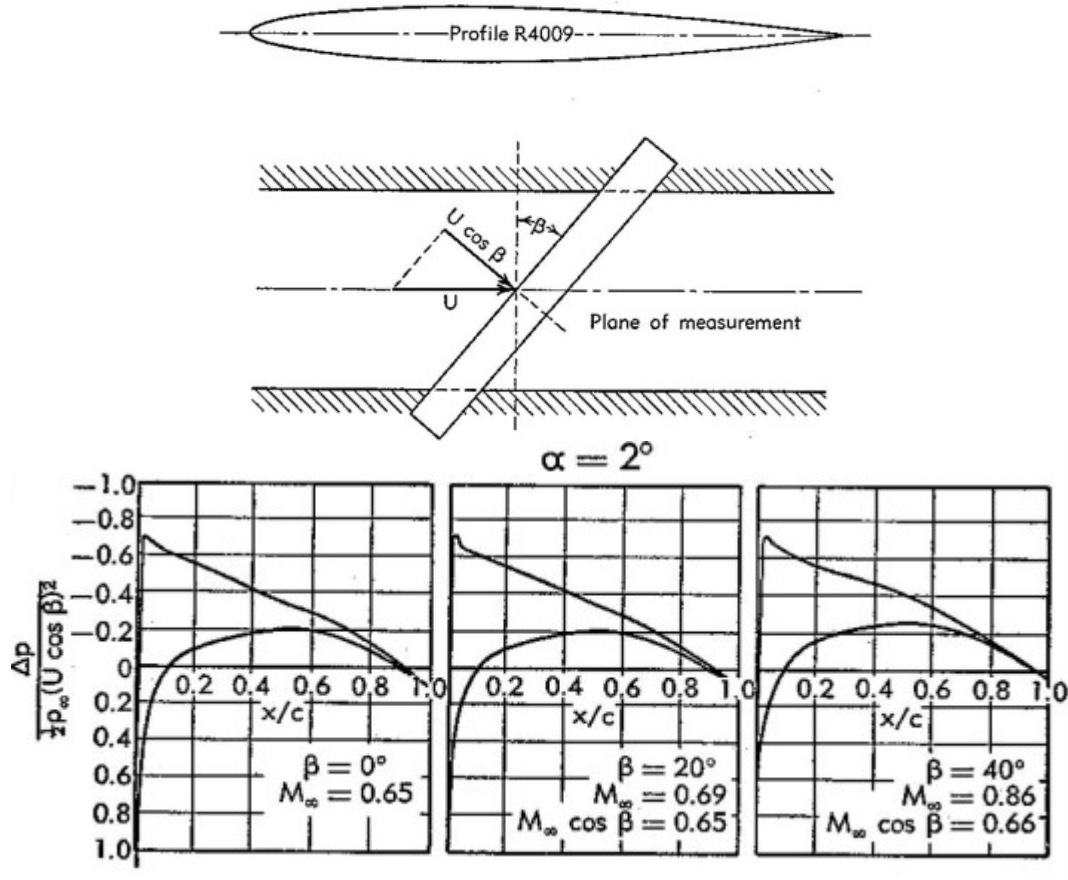


FIGURE 9 – R.T. Jones rework of the Braunschweig Swept Wing Wind Tunnel Data [26]

D. American Consensus Leveraging Simple Sweep Theory

As we note in the introduction to this paper, John Anderson is a strong advocate of Simple Sweep Theory. Examples of it are found in his *Introduction to Flight* as well as his *Fundamentals of Aerodynamics* textbooks.[5][6]

Drela [30] describes only the “infinite swept wing” for both low-speed and compressible flow conditions.

Kuethe & Schetzer [31] follow R.T. Jones and describe only the “infinite yawed wing.” They hold that the Mach number normal to the leading edge is the sole arbiter of shock formation: if the Mach number normal to the leading edge “is less than the velocity of sound, the wing, in effect, is moving subsonically and the wave drag associated with supersonic flow is absent.”[31] Kuethe & Chow [32] reiterate R.T. Jones’ [29] description of the “infinite yawed wing;” they present a re-plot of the Braunschweig wind tunnel data and argue that it shows that the pressures are solely a function of the flow component normal-to-the-leading-edge.

Confusingly, elsewhere in his otherwise excellent book, McLean [4] states that the pressure coefficient of the three-dimensional wing is the two-dimensional data transformed by $(\cos \varphi)^2$: $Cp_{3D} = Cp_{2D}(\cos \varphi)^2$. This would imply that he condones Simple Sweep Theory, which contradicts his other writings found elsewhere in the same book.

E. American Authors Avoiding a Definitive Explanation of Sweep Effects

Ashley & Landahl [33] largely avoid any discussion of wing sweep briefly introduce the concept of subsonic and supersonic-leading-edges before moving on to more complex, mathematical models keyed only to freestream Mach number conditions.

Brandt, et. al [34] avoids any deep theoretical explanation of sweep effects; these authors limit themselves to presenting an empirical equation where the Critical Mach Number is a function of the cosine of the leading-edge sweep raised to the 0.6 power; e.g., $\cos^{0.6}(\varphi_{LE})$. This is not congruent with Simple Sweep Theory. Circumstances imply that the 0.6 exponent is an empirically derived value; however, Brandt does not cite the source of his unique formulation.

Bertin & Cummings [35] address swept wings all too briefly in their book *Aerodynamics for Engineers*. They reproduce data from the original German test series and only discuss the qualitative effects of sweep. They neither discuss how to design an aircraft to a given sweep angle nor do they specify a criteria to predict incipient shock formation on the swept wing.

Similarly, Cummings, Mason, Morton & McDaniel [36] do not provide a comprehensive discussion of sweep effects. They hint at the universality of Simple Sweep Theory when they present the Swept Wing Korn Equation, an empirical relationship between design CL , t/c , and the *Critical Mach Number*. [37] Even then, they state that this is an “approximation” which they “calibrate” using Boeing transport category geometric and flight performance data.

Schairer in his memoirs, wrote that when Boeing “first tested swept wings, we found that the lift and drag Critical Mach Numbers were delayed by sweep by only **two-thirds** of what was predicted by Busemann’s cosine theory.” [17] He continues by stating that people at Boeing “assumed that the problem lay in flow over the inner portion of the wing” or due to interference between the wing and fuselage. As of his writing in 1980, Schairer notes that “this cut-and-try tailoring approach to wing aerodynamics has been necessary on every jet ... Boeing has built.” [17]

F. British Approaches to Swept Wing Aerodynamics

The Germans did not have a monopoly on the interest in compressible flows. Before his premature death in 1934, Hermann Glauert, the principal scientific officer of the Royal Aircraft Establishment at Farnborough, was deeply immersed in the study of compressible flows.

An occasionally cited, but essential observation is found in ARC R&M 2330. [12] Here, Bickley focused on the concept of the “Critical Mach Number” as applied to compressible fluid flows. Below the Critical Mach Number, exact solutions to steady irrotational isentropic flow can be found. Similarly, exact solutions to supersonic flow can be found. What challenged Bickley was to identify the physical criteria associated with the formation of the sharp discontinuities in flow properties we know as a “shock wave.” While the R&M dates from 1946, Bickley states that in earlier unpublished work he realized that “fluid acceleration occurs in the direction of the (negative) pressure gradient.” [12] And “if the component of the fluid velocity in this direction were greater than the local speed of sound, then pressure changes could not be propagated upstream, and the flow would not adjust itself to downstream conditions.” [12] He, therefore, claims that the sweep angle of “isobar lines” of the flow field is the important governing parameter, not the geometric sweep of the leading edge that other authors use. Bickley holds that the sweep of the isobar line applies equally to the flow across an oblique shock wave as to the flow over a finite wing.

Bickley’s [12] approach forms the basis of the Küchemann Critical Pressure Coefficient equation: [11]

$$C_p^* = \frac{2}{\gamma M_\infty^2} \left\{ \left(\frac{2}{\gamma+1} \right)^{\frac{\gamma}{\gamma-1}} \left(1 + \frac{\gamma-1}{2} M_\infty^2 (\cos \varphi)^2 \right)^{\frac{\gamma}{\gamma-1}} - 1 \right\} \quad (5)$$

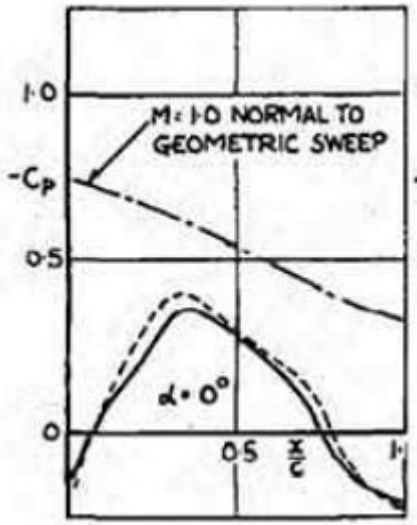


FIGURE 10 - Finite Wing pressure data from Newby during the development of the Phase II Avro Vulcan Wing [8][13]

where ϕ is the sweep of the “isobar line” of interest.[11] The reader should note that the freestream Mach number is found in both the numerator and denominator, yet the sweep of the isobar line only transforms the Mach number in the numerator; thus, the denominator remains dependent upon the freestream Mach number, without the sweep transformation, similar to Schlichting. Unlike a literal application of Simple Sweep Theory, this method captures elements of the magnitude of the freestream as well as the leading-edge flow.

Küchemann notes that “Bickley’s criterion has been confirmed [by] many experiments.”[11]

An example of the application of Bickley’s rules to the design of the target upper surface pressure distribution of a wing may be seen in Newby.[8][13] This work, documenting the development of the highly three-dimensional geometry of the Phase II Avro Vulcan wing, shows how the target peak suction pressure coefficient varied from leading edge (where the sweep was the highest) to the trailing edge (which was essentially unswept). As we enter the Küchemann C_p^* equation with the same freestream Mach number but increase the sweep the maximum permissible sub-critical C_p grows more negative; see FIGURE 10.

With this insight, many production British aircraft, which otherwise predate advanced computational aerodynamics tools, were able to feature highly three-dimensional geometry. Obert showcases many wings in his text, the *Aerodynamic Design of Transport Aircraft*. [38] Beyond the Phase II Avro Vulcan, the Handley Page Victor, the Vickers VC-10 and the BAC 1-111 all embody the idea that a high-performance transonic swept wing must have strong spanwise variation in airfoil shape in order to render the smooth isobar patterns needed to prevent premature shock wave formation. In hindsight, we note that these aircraft were all designed during the era where Schairer at Boeing recalled the need for “cut-and-try tailoring” of airfoils.[17]

G. Kirkman & Takahashi’s confirmation that Küchemann was correct.

Recent computational work by Armenta, Plaban & Takahashi [39] as well Kurus, Takahashi & Perez [40] reconfirms this truth; that the critical pressure coefficient which controls incipient shock formation is keyed to the sweep of the isobar pattern which develops over an aircraft.

In *Understanding Aerodynamics*, McLean agrees with Küchemann and states that the critical conditions are dependent upon the Mach number perpendicular to the isobars of the wing.[4]

Thus, we believe that Küchemann’s equation effectively predicts incipient sonic flow. The incorporation of this method into the potential-flow based wing design process can therefore enable the development of new rapid physics based geometric synthesis tools.

At the same time, many aerodynamicists continue to rely upon 2-D Simple Sweep Theory.

In this paper, our new work, using modern volume grid CFD to revisit the original LFA Braunschweig data set which was used to “prove” (or disprove) the quantitative application of Simple Sweep Theory.

III. History of the Wind Tunnel Test Facilities

By the Second World War, the German Aeronautical Research Institute or the *Deutsche Luftfahrtforschungsanstalt* (LFA-Braunschweig) began construction in 1935 and commenced operation by 1936. The LFA facility was largely hidden in dense forestry or were well camouflaged and the site remained so secretive that its precise location was not discovered by the Allies during the war; see FIGURE 11.[41]



FIGURE 11 – LFA Wind Tunnel Complex near Braunschweig, Germany [41]

This facility was mainly responsible for airframe, aeroengine and aircraft weapons testing. Prominent manufacturers included BMW, Focke Wulf, Messerschmitt, and Junkers; see FIGURE 12. The Bf-109 and Fw-190 would consistently assert their formidable air combat capabilities throughout the Second World War. The LFA facility would be critical in the development of turbojet powered aircraft such as the Messerschmitt 262 and the Arado 234. Because turbojet powered aircraft can operate at much higher altitudes and airspeed than piston aircraft, their design relied upon the swept wings stemming from Busemann's work.

By 1945, there were at least a total of eight subsonic, transonic and supersonic wind tunnels in operation or calibration phases for research and project support.[41] The low-speed wind tunnel, A-1, was utilized for basic research requirements such as laminar flow characteristics and capturing low speed lift, drag, pitching and control moments and would achieve a speed of 55 m/s.[41] The high subsonic wind tunnel, A-2, could achieve Mach 0.82 (300-m/s) and was utilized to investigate drag rise. One key discovery made at the A-2 wind tunnel was that a designer could delay the drag rise by shifting an airfoil's maximum thickness point aft of an airfoil or nacelles.[41] Both A-1 and A-2 had small test sections being 2.5-2.8 meters in diameter. Wind Tunnel A-3 would only achieve a speed of 95 m/s but had a much larger 8-meter diameter test section and was utilized for stability and control tests on submarine models and full-size torpedoes.[41] A-3 would also serve as a test bed for running piston and jet engines for firms like the aforementioned BMW, Junkers and Arado where testing of wing aft and forward sweep was undertaken; see FIGURE 13.[41]

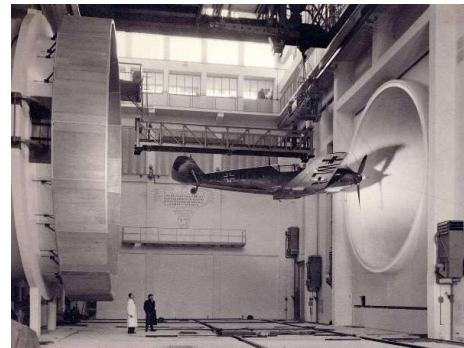


FIGURE 12 – Bf. 109 Wind Tunnel Test [42]

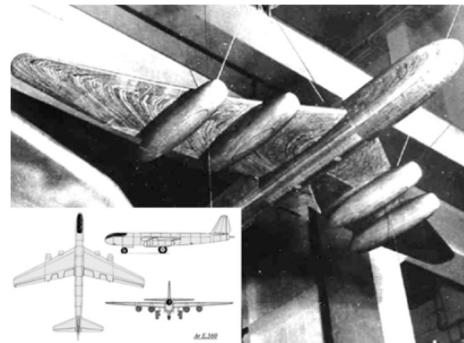


FIGURE 13 – Ar-234 Swept Wing Model in A-3 [41]

By 1948 all wind tunnels at LFA were dismantled by the British Military Government. This included Busemann's Transonic A-9a/b wind tunnel. The A-9 building housed a pair of supersonic wind tunnels capable of reaching a speed of Mach 1.5 with a test section diameter of only 80 cm^2 . [42] Much of Busemann's transonic wind tunnel test and pertinent information regarding his swept wing analysis at LFA are scattered amongst various NACA reports and other institutional publications. Busemann's Theory on wing sweep suggested that a swept wing of angle ϕ (also noted as Λ or β in various works) acts like an infinite span wing yawed at angle ϕ to the wind. [40] However, Busemann's Simple Sweep Theory assumes that the spanwise flow has no major impact on airfoil and subsequently wing performance. This is a shortcoming in his theory, yet this served as a basis that famous authors such as Küchemann, [11] Schlichting [43] and Anderson [6] would build upon. Additional insights on the experiments conducted at LFA are described by Lippisch & Beuschausen, in the 1947 NACA TN 1115. [44]

IV. Details of the LFA Wind Tunnel Used to Collect the Yawed Wing Dataset

Lippisch notes that the wing is situated in a $0.4 \times 0.4\text{ m}^2$ space within the wind tunnel where the wing spans the width of the test section. The wing is analyzed at sweep angles of 0° , 20° and 40° in combination of angles of attack ranging from 0° to 3° . [44] Since this cross-sectional area is much smaller than the identified wind tunnels, A-1, A-2, and A-3, and seemingly larger than the test sectional area of A-9, this provides insight on the sizes of the remaining four wind tunnels used at LFA which could have been used to conduct these experiments.

To have a more faithful remodeling of wind tunnel setup and capture similar pressure coefficient distributions as seen in FIGURE 7, it is critical to examine an airfoil that most closely estimates the R4009; see FIGURE 14. Lippisch states that the R4009 airfoil was a "hyperbolic profile" with a maximum thickness located 40% aft. [44] History has not preserved the ordinates for the R4009 section. Since Harry E. Murray in NACA TN 1739 [45] references TM 1115 [46] and mentions that the R4009 is best approximated by a NACA 0012-64 airfoil, we use the 0012-64 as our basis to reverse engineer Busemann's experiment with modern CFD.

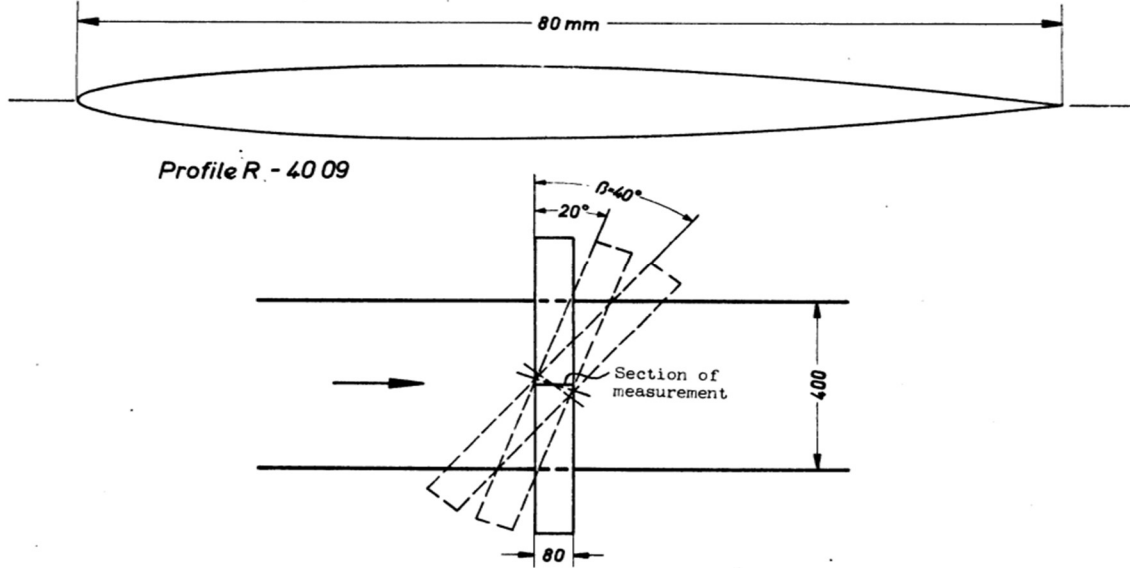


FIGURE 14 – Sketch of the Wind Tunnel Model from Ref [45]

V. Modern CFD replicating LFA Wind Tunnel Test Geometry

Coupled with the insights gained from Jones, [24][26][27][28][29] Lippisch & Beuschausen, [44] and Murray, [45] we can recreate the famous swept wing experiments using ANSYS Fluent.[47]

A. ANSYS Grid Geometries and Meshing Strategies

We model our volume grid in a fashion that remains faithful to the wind tunnel drawing as shown in FIGURE 7.[21] This means that the freestream airflow will run parallel to the wind tunnel walls, as opposed to offsetting the volume grid respective to the sweep angle. As the wing is increasingly swept, the freestream Mach will be adjusted accordingly to satisfy the condition in which the incident airflow which runs perpendicular to the leading edge remains at approximately Mach 0.65. As mentioned earlier, as the sweep angle of the wing increases, the chord of the wing increases with the inverse of the cosine of the sweep angle. It should be noted that the following results are performed with a scaled-up version of the wing and wind tunnel test section due to inherent limitations in using the student version of ANSYS Fluent. Additionally, another significant limiter in using the student version of ANSYS is the cell count. In ANSYS 2024 R2, the Fluent cell limit for the student version is 1,048,576 (2^{20}) cells and 16 cores of processing capability

This is largely sufficient for many 2-D studies including Large Eddy Simulations or conducting high fidelity RSTM (Reynolds Stress Transport Model) studies. However, this is generally coarse for most 3-D CFD analyses. The full version of ANSYS does not impose such limitations on cell count in contrast.[48] As a result, any numerical data generated may inherently contain significant errors and should be interpreted primarily to identify general trends rather than precise values. Consequently, the subsequent analysis will focus largely on qualitative aspects. This includes examining isobar distributions and shock formations, where the color gradient will serve as a reference for areas of very low and high pressures and airspeed, rather than providing definitive measurements for the case being studied.

In ANSYS, there are predominantly two geometries that are used in forming the mesh. In a 2-D analysis, this will either be quadrilaterals (QUAD4) or triangles (TRI3) which contains 4 nodes and 3 nodes of computation respectively.[49] Given the extra node and a geometrically stable shape, the QUAD4 mesh will allow for better control of the mesh as well as lower error and increase the probability of convergence at the cost of being more computationally intensive.[50][51]

In a 3-D space the meshing geometries will be either hexahedral (HEX8) or tetrahedral (TET4) or a combination of the two.[51] The choice between a hexahedral or tetrahedral geometry is a widely debated topic. Many practitioners deem hexahedral is generally considered better because it consists of 6 faces per element and 8 nodes of computation thus creating a more stable geometry. The tetrahedral mesh will have 4 faces and 4 nodes meaning that on average the overall computational fidelity will be less for the same element size. This makes hexahedral meshing geometries more computationally economic as 6 tetrahedrons are needed to resolve each hexahedron.[52] This also suggests that a hexahedral mesh geometry generates a more refined model that will converge to a solution sooner.[53] However, because hexahedral meshes have 6 degrees of freedom to resolve, mesh generation, especially around complex geometry like airfoils, is difficult and very time intensive. On the other hand, tetrahedral geometry is not as structured compared to the cuboidal hexahedral geometry.

Since the tetrahedral geometry has more flexible generation conditions, it appears to be optimal in capturing incident flow and is able to resolve complex geometry such as the leading edge; see FIGURE 15. FIGURE 15 also shows that for a tetrahedral geometry, the meshing algorithm will overlay other tetrahedral geometries and enable node sharing while incorporating an inherent difference in shape also as deemed appropriate (isosceles/equilateral/etc.). This

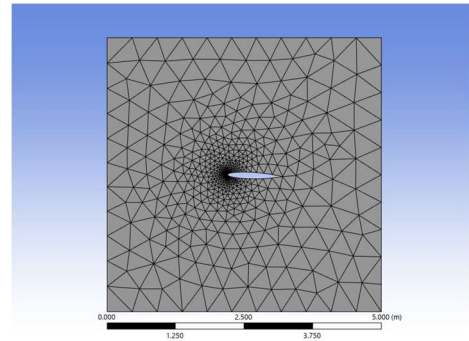


FIGURE 15 – TRI3/Tetrahedral Mesh

serves as reasoning for why tetrahedral meshes will usually have a far smaller node count compared to its element count.

This suggests that when comparing control volumes with approximately the same element count, the difference in node counts between the two meshing geometries may suggest faster computation for the tetrahedral mesh at the cost of accuracy and likely convergence as well. Conversely, populating the control volume with more tetrahedral elements to match the same node count to improve the accuracy of the analysis can be performed but this means that for a tetrahedral mesh, the solver will have to solve 5-6 times more elements which will substantially increase solver time.

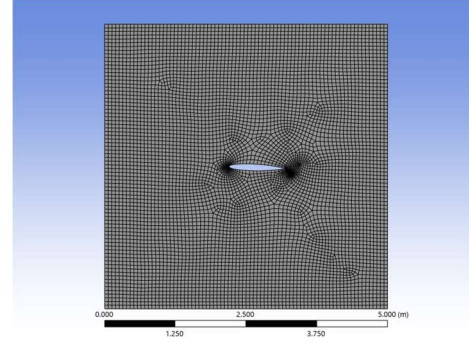


FIGURE 16 – QUAD4/Hexahedral Mesh

With this understanding, a QUAD4 geometry for 2-D cases and a HEX8 geometry for 3-D cases is applied whenever possible; see FIGURE 16. Furthermore, given the cell limitation that is associated with the student version of ANSYS, and the inherent numerical error associated with it, the following results utilize a grid density that adequately captures the flow behavior and shock patterns while mitigating time intensive processes during mesh generation and computation.

B. Initial 2-D Study of the 3-D Airfoil

We conduct a preliminary analysis in the 2-D reference frame to isolate the airfoil's baseline aerodynamic performance **without** sweep effects. Considering the limitations with mesh sizes, the NACA 0012-64 is recreated and analyzed in ANSYS Fluent and scaled to have a 1-meter chord length situated in a 5-meter tall and 5-meter spanning test section thereby scaling up the original 80-millimeter chord length and the $0.4 \times 0.4 \text{ m}^2$ test section. In later sections of this paper, when examining the 3-D cases, the length of the tunnel will vary dependent on the sweep angle being studied to maintain sufficient distance between the inlet and outlet of the wind tunnel to maintain proper upstream flow characteristics and to prevent potential back flow that may occur while balancing the limited cell density. In performing this action, mesh generation for the control volume has shown to be more stable and does not result in either a failed or obsolete grid and is relatively time efficient. The analysis is performed using air modeled as an ideal gas, with a default density of 1.176655 kg/m^3 . All cases are investigated using a density-based solver under inviscid, steady-state conditions. Utilization of a density-based solver is necessary in order capture compressibility effects.[54] For the following 2-D cases, refer to TABLE 1 for the associated airfoil geometries, freestream Mach number and grid densities.

TABLE 1 - 2-D Unswept Cases

Case #	M_∞	Reference Velocity (m/s)	Chord Length (m)	Element Size (m)	Element Count	Mesh Geometry	3-D Test Sweep (φ)	2-D t/c representing 12% Normal to LE (%)	Küchemann C_p^* (unswept, $\varphi = 0^\circ$)
1	0.65	225.6	1	0.05	10297	QUAD4	0°	12.0%	-1.009
2	0.69	239.5	1.063	0.05	10261	QUAD4	20°	11.3%	-0.821
3	0.86	298.5	1.305	0.05	10326	QUAD4	40°	9.2%	-0.278

The wing is set at 2° angle of attack (α) and analyzed in pure 2-D flow (i.e., unswept) at the freestream Mach numbers used in the famous swept wing Busemann study; $M_\infty=0.65, 0.69$ and 0.86 .

FIGURES 17-19 highlight Mach contours for each of the cases from Table 1, above.

For the NACA 0012-64 airfoil, we see in FIGURE 17 that the critical Mach (M_{cr}) occurs just above $M_\infty=0.65$ as the flow over the top surface of the airfoil reaches a peak of $M=0.981$ near the leading edge.

In the absence of sweep, when the freestream flow is set to $M_\infty=0.69$, the wing passes its Critical Mach Number. Flow over the top surface of the airfoil reaches $M=1.07$ near the leading edge; see FIGURE 18.

As we further increase the freestream flow to $M_\infty=0.86$, the develops fully supersonic flow over both its upper and lower surfaces; see FIGURE 19. Peak flow velocity reaches $M=1.44$ and a clearly defined shock wave forms at the trailing edge. This shock extends to the top and bottom of the control volume and “chokes” the wind tunnel. This serves as a basis for what this airfoil’s two-dimensional aerodynamics is.

The formation of either minor perturbation shocks or stronger shocks typically occurs as the pressure distribution subsequently drops below the critical pressure coefficient. Additionally, the gradient of the pressure distribution after it falls below the critical pressure serves as an indication of the shock’s strength.

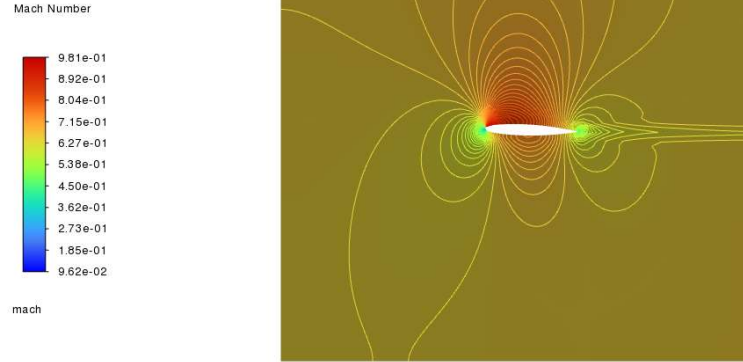


FIGURE 17 - 2-D Mach Contour. $\varphi = 0^\circ$; $t/c=12.0\%$; $M_\infty = 0.65$.

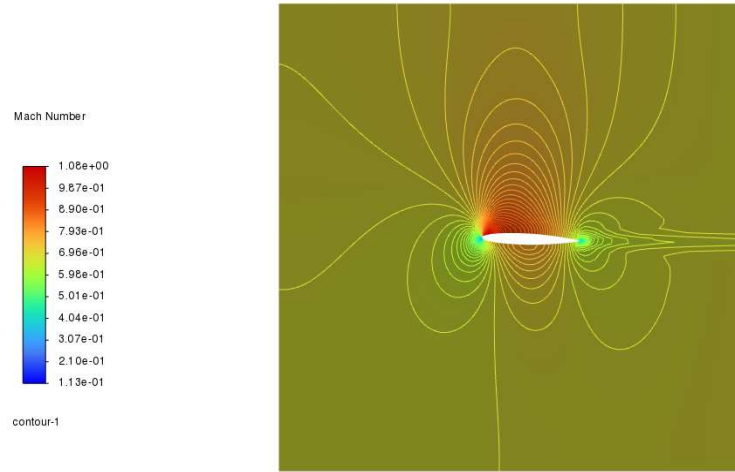


FIGURE 18 - 2-D Mach Contour. $\varphi = 0^\circ$; $t/c=11.3\%$; $M_\infty = 0.65$.

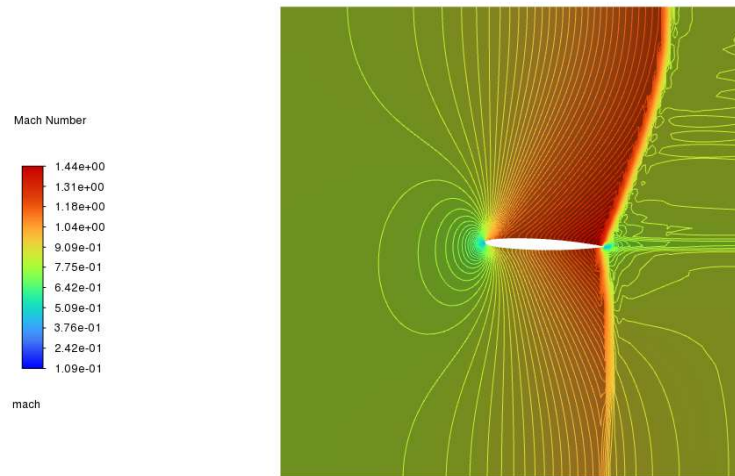


FIGURE 19 - 2-D Mach Contour. $\varphi = 0^\circ$; $t/c=9.2\%$; $M_\infty = 0.65$.

When examining the pressure coefficient plots in FIGURES 20-22 over the chord length, we observe that the trends outlined by Küchemann's equation (5) for 3-D swept wings conceptually apply even to 2-D cases. For the unswept, 2-D section ($\varphi = 0^\circ$) studied here, C_p^* varies with free-stream Mach number; see TABLE 1.

When examining FIGURE 20, the minimum pressure coefficient ($C_{p_{min}}$) is just below Küchemann's C_p^* (-1.009) which suggests that the flow goes with a slightly subcritical regime meaning that the local Mach number for the upper surface is less than 1 but is approaching it closely.

Similarly, when looking at FIGURE 21, $C_p^* = -0.821$, the shockwave appears to form at approximately 25% chord on the upper surface only. This demonstrates the flow accelerating and reaching supersonic speeds before decelerating rapidly due to the shock. Due to the low grid density, the resolution of this shock is not as sharp or distinct, making it challenging to visualize and accurately capture its exact location and characteristics.

Now when moving onto FIGURE 22, where $C_p^* = -0.278$, the entire upper surface of the airfoil is supercritical. Additionally, the lower surface is also supercritical aft of ~15% chord all the way to the trailing edge. Given these conditions, it is not surprising that this airfoil exhibits a shockwave anchored to the trailing edge. Thus, we confirm that the unswept Küchemann C_p^* equation accurately predicts sonic flow.

With this understanding of 2-D airfoil flow behaviors, when incorporating sweep, it is evident that 3-D flow effects play a crucial role in determining the overall wing performance. Swept wings introduce spanwise flow and pressure relief components that alter the pressure distribution and delays the onset of shock waves at high speeds. This spanwise pressure relief affects how lift is generated and redistributed along the wing.-Consequently, the aerodynamic characteristics, such as drag and lift, must be reevaluated in the context of 3-D flow behavior. Understanding these effects is essential for optimizing wing designs, especially for high-speed applications where sweep can improve performance by mitigating wave drag and improving transonic and supersonic handling.

We should note an important trend with these 2-D solutions. In the absence of wing sweep and 3-D effects, the computed pressure coefficients are **completely unlike** those showcased in FIGURE 15 (prior section). According to the other authors, these pressure distributions should be identical (or nearly identical).

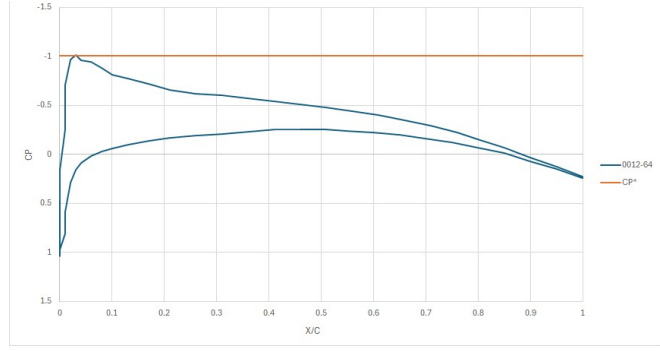


FIGURE 20 – 2-D C_p Distribution. $\varphi = 0^\circ$; $t/c=12.0\%$; $M_\infty = 0.65$.

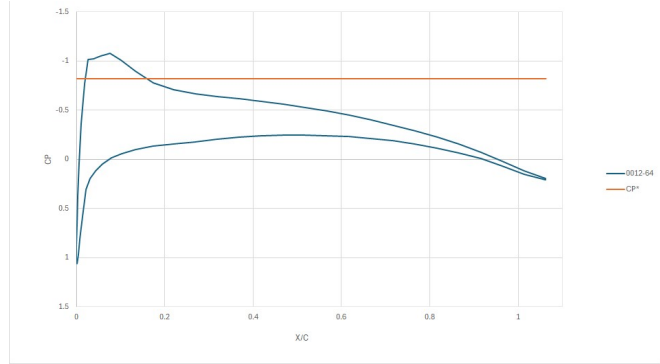


FIGURE 21 - 2-D C_p Distribution $\varphi = 0^\circ$; $t/c=11.3\%$; $M_\infty = 0.69$.

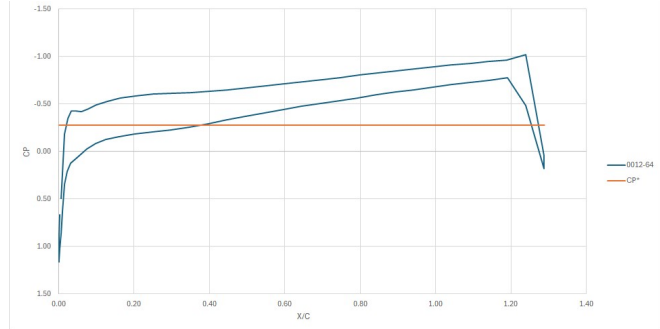


FIGURE 22 - 2-D C_p Distribution. $\varphi = 0^\circ$; $t/c=9.2\%$; $M_\infty = 0.86$.

In the next section, we will look at the NACA 0012-64 airfoil in the unswept, 20° and 40° sweep cases set at 2° angle of attack as investigated by Busemann's wind tunnel tests. We will also investigate alternative angle of attack values to understand how the operating parameter will contribute to the severity of the shock formation. This includes achieving incipient criticality when the flow approaches the Critical Mach Number as well as the strength and prominence of the shock itself.

C. Replicating the Geometry of the Wind tunnel Spanning Unswept and Swept Wings – 3-D

Given that sweep is inherently a three-dimensional design feature, we will now examine the flow over the “wind tunnel” spanning swept airfoil.

Since Busemann's original setup is being viewed in the wind axis, as mentioned previously, the chord length and freestream Mach increases by Eqn. 6 and 7, respectively:

$$M_{\infty}(\varphi) = \frac{M_{\infty}}{\cos(\varphi)} \quad (6)$$

$$c_{wind-axis}(\varphi) = \frac{c}{\cos(\varphi)} \quad (7)$$

The associated wind tunnel setups are shown in FIGURES 23-25. If sweep works as claimed, the chordwise pressure distributions along the centerline airfoil should be essentially identical so long as we keep the Mach number normal to the leading edge constant. Unlike the 2-D case, when we raise the freestream Mach is raised to Mach 0.86, the airfoil should remain shock free; the wind tunnel will not choke.

The CFD setup when conducting 3-D analysis on Busemann's wing remains akin to the previously discussed 2-D analysis. Since the freestream Mach, M_{∞} , is increased to transonic speeds, it is important to utilize a density-based solver to capture compressibility effects accurately. In addition, just as is in the 2-D case studies, an inviscid and steady state flow quality is imposed. Doing so will reduce the wall effects on the pressure distribution by enforcing a slip condition on the walls of the volume grid. Enough iterations will be assigned to allow the simulation to converge to a solution. Pressure coefficient measurements will be taken along the midspan to minimize capturing wall effects. This also coincides with the spanwise location that Busemann defines as the “plane of measurement” in FIGURE 9.

For the following 3-D cases, refer to TABLE 2 for the associated airfoil geometries, freestream Mach number and grid densities.

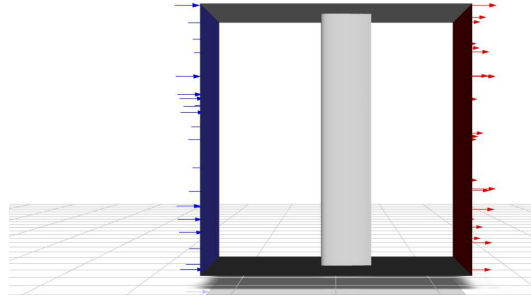


FIGURE 23 – 3-D Model. $\varphi = 0^\circ$; $t/c=12.0\%$; $M_{\infty} = 0.65$.

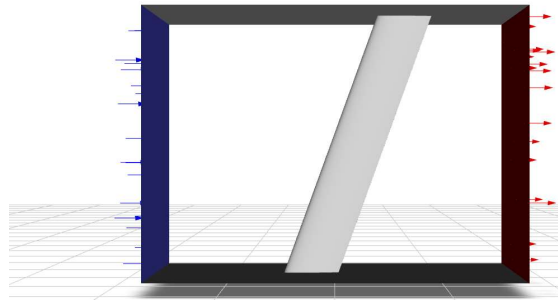


FIGURE 24 – 3-D Model. $\varphi = 20^\circ$; $t/c=11.3\%$; $M_{\infty} = 0.69$.

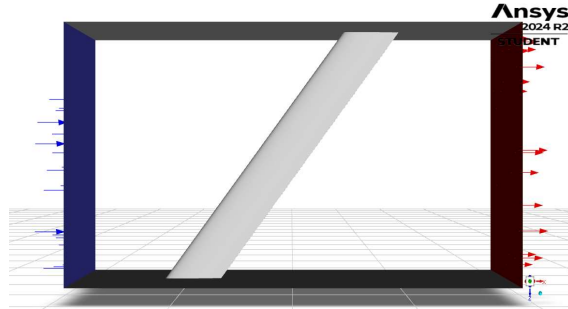


FIGURE 25 – 3-D Model. $\varphi = 40^\circ$; $t/c=9.2\%$; $M_{\infty} = 0.86$.

TABLE 2 - 3-D Cases

Case #	M_∞	Reference Velocity (m/s)	Chord Length (m)	Element Size (m)	Element Count	Mesh Geometry	3-D Test Sweep (φ)	2-D t/c (%)	M_{LE}	Küchemann C_p^* (following sweep)
1	0.65	225.6	1	0.38	544185	HEX8	0°	12.0%	0.65	-1.009
2	0.69	239.5	1.063	0.251	542052	HEX8	20°	11.3%	0.65	-0.898
3	0.86	298.5	1.305	0.95	758471	TET4	40°	9.2%	0.66	-0.566

As mentioned, a QUAD4 or HEX8 mesh is the preferred meshing geometry, however as the sweep angle is increased, meshing the setup with a hexahedral geometry became increasingly difficult. Forcing ANSYS to mesh using an “Automatic (PrimeMesh)” method to generate a hexahedral geometry with an aggressive triangle reduction criterion did not yield the desired result. Alternative meshing methods such as “Sweep,” “MultiZone” and “Cartesian” methods would fail to generate. For this reason, the 40° sweep cases are analyzed using a tetrahedral geometry while maintaining roughly the same mesh count as is performed with the unswept and 20° swept cases. However, given that the analysis conducted given the node limitation was to be strictly qualitative, we feel that the solutions are accurate enough to validate Busemann’s claims.

D. How “Transonic” is the original dataset?

As the 0012-64 wing is increasingly swept, it is apparent in FIGURES 26-28 that the freestream accelerates to approximately the speed of sound along the midspan, which is congruent to the aforementioned “plane of measurement”. Additionally, as the sweep is increased, the strength of perturbations and shocks increases near the leading-edge of the down-wind wing section. This is clearly visible from the Mach contours: in FIGURE 26, the flow is entirely subcritical; in FIGURES 27 and 28, the flow accelerates considerably past the sonic near the right-hand-edge of the control-volume (representing the wind tunnel-wall).

FIGURE 28 shows how the flow in the virtual wind tunnel begins to choke even though the wing is swept. As we move across the wing, the surface lines of constant Mach number are in no way aligned with the wing sweep; there is substantial three-dimensionality. All across the span, flow over much of the upper surface is supersonic. As we approach the down-wind wall, we see a sharp deceleration from locally supersonic to subsonic flow – this is a shock wave which has formed normal to the inbound

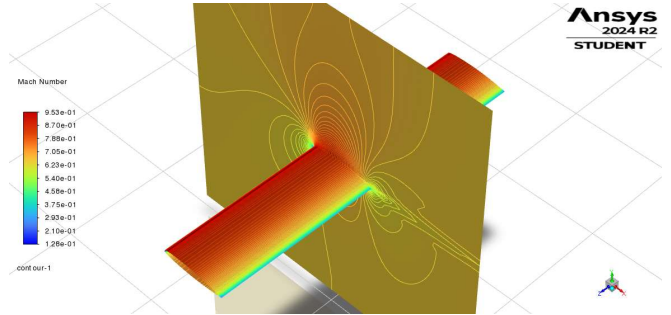


FIGURE 26 – 3-D Mach Contours. $\varphi = 0^\circ$; $t/c=12.0\%$ $M_\infty = 0.65$.

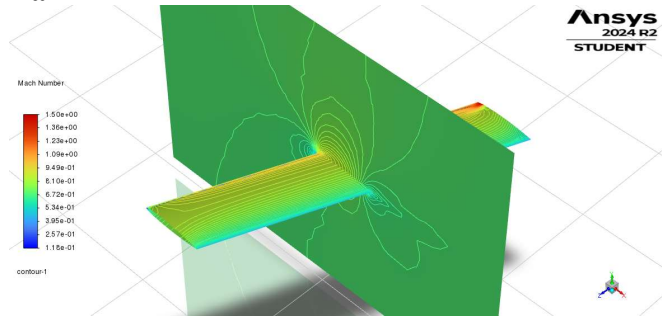


FIGURE 27 – 3-D Mach Contours. $\varphi = 20^\circ$; $t/c=11.3\%$ $M_\infty = 0.69$.

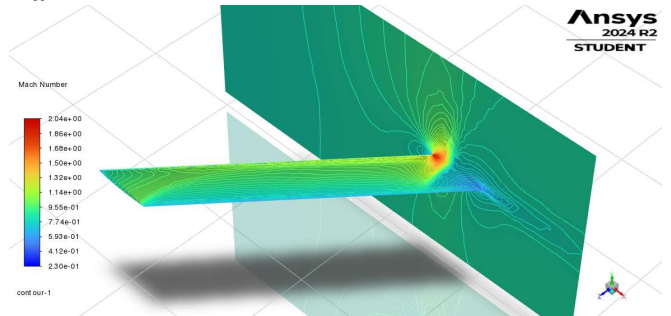


FIGURE 28 - 3-D Mach Contours. $\varphi = 40^\circ$; $t/c=9.2\%$ $M_\infty = 0.86$.

flow (and is completely decorrelated with the wing sweep). As expected, this is in stark contrast to the 2-D cases discussed previously: at Mach 0.86 the tunnel downstream of the wing was choked by a shock that formed at the trailing-edge. Furthermore, at the wing root, the airspeed accelerates to roughly sonic speeds by $\sim 85\%$ chord aft of the leading edge. Moving down the span from this point, the airflow accelerates to sonic speeds sooner and sooner until the midspan whereafter the flow speed, as mentioned, increases into the supersonic regime.

Thus, we see that the original Busemann dataset – which was ostensibly shock free – is much more transonic than anyone suspected. Shock waves form at some spanwise location for both these cases.

E. What amount of Three-Dimensionality is implied in the original test data?

It is also important to examine the pressure coefficient contour lines as well; these “isobar” plots are shown as FIGURES 29-31.

In FIGURE 29, the wind tunnel spanning unswept, subcritical wing shows a smooth, almost 2-D, isobar pattern.

As we add sweep, things change markedly. Unlike the 3-D unswept case in FIGURE 29 where the contour lines run straight, moving from left to right, the swept cases (FIGURES 30 and 31) show substantial spanwise variation in the chordwise pressure distributions. While the isobar lines are fairly straight at the centerline of the model, they grow increasingly distorted as we approach the virtual “wind tunnel walls.” In FIGURES 30 and 31, the formation of shock waves near the leading-edge of the down-wind wing section is very clear. In FIGURE 30, the pressure coefficients jump from $C_p < -1.6$ to $C_p \sim -0.6$ over a very short distance; recall K  chemann’s C_p^* for this geometry is -1.0831 . Similarly, in FIGURE 31, the pressure coefficients jump from $C_p < -0.6$ to $C_p \sim -0.2$ over a very short distance; recall K  chemann’s C_p^* for this geometry is -0.566 .

Given the strong spanwise variance in airspeed and pressure coefficients, it is unreasonable to characterize the analysis in a quasi-2-D perspective. In reference to FIGURES 28 and 31, in a quasi 2-D analysis, the isobars would likely look straight at the wing root moving towards the midspan and will show distortion due to the strong adverse pressure gradient near the wing tip. This would result in severely skewed performance because of an oversimplification of the complexities of three-dimensional flow. This will consequently pollute the pressure coefficient data which will show that certain areas along the wing are subcritical when it may very well not be. At 40° of sweep, the Mach contour in FIGURE 28 shows the airspeed reaching near criticality as

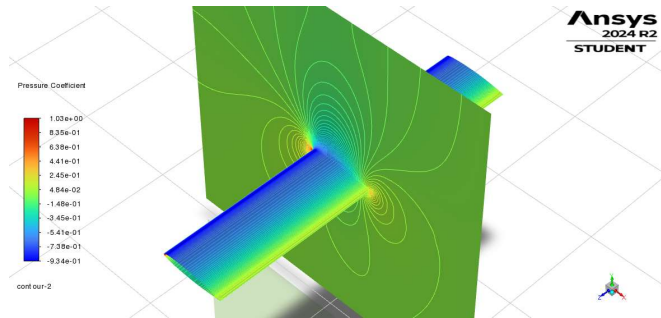


FIGURE 29 – 3-D C_p Distribution. $\phi = 0^\circ$; $t/c=12.0\%$
 $M_\infty = 0.65$.

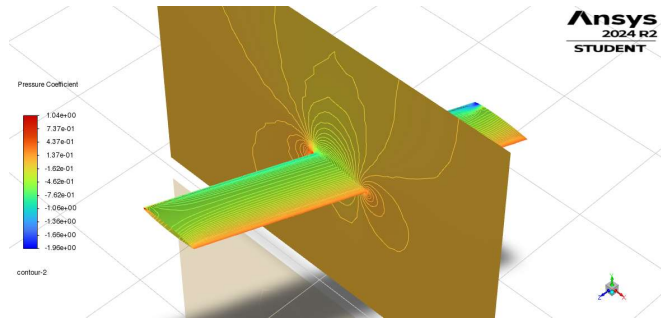


FIGURE 30 – 3-D C_p Distribution. $\phi = 20^\circ$; $t/c=11.3\%$;
 $M_\infty = 0.69$.

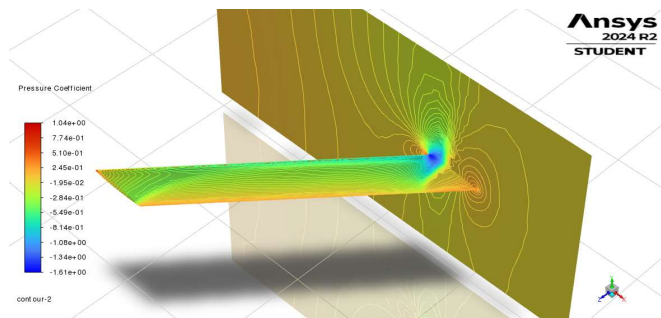


FIGURE 31 - 3-D C_p Distribution. $\phi = 40^\circ$; $t/c=9.2\%$;
 $M_\infty = 0.86$.

mentioned previously which would not be captured in a quasi-2-D analysis. This means that generally, pressure coefficients would appear qualitatively smoother over the span and quantitatively deflated and underpredict shock strength and how prominent the shock is on the wing.

F. Comparing Modern CFD to Classic Test Data

Here we compare the pressure coefficients along the mid-span of our CFD models with the experiments conducted at the LFA wind tunnels. It should be noted that we approximate the R4009 airfoil with the NACA 0012-64, we cannot expect a perfect match. We begin and overlay the pressure coefficients on top of Busemann's original data; see FIGURES 32-34.

It is apparent that there is noticeable variation in the upper surfaces C_p distribution due to grid density. In FIGURE 32, the $\sim 1,000,000$ cell mesh solution closely matches test data; there is some error obvious with the $\sim 540,000$ cell mesh. However, when examining FIGURE 33, there is virtually no difference between the coarse and fine mesh. This suggests that the coarser mesh of $\sim 540,000$ cells was sufficient to accurately capture the pressure distribution. We also note that Küchemann predicts the 3-D upper surface C_p be slightly subcritical; whereas the 2-D solution with transformed t/c and M_∞ was clearly supercritical; recall FIGURE 21.

FIGURE 34 is unique in that it chokes the tunnel; see also FIGURE 28. This is a crucial aspect as choking the tunnel disturbs the flow everywhere.

At Mach 0.86, where the flow over the top of the wing is supersonic and a normal shock wave is forming across the “virtual test section,” compressibility effects dominate. This prevents the CFD solver from fully converging. If we allow the solver to continue iterating in the attempt to better converge, the residuals drastically rise.

When examining a Mach contour like those in FIGURES 26-27, this divergence is signified by a nucleation point at a seemingly arbitrary point along the model. In instances where this occurs, running the solver for enough iterations just prior to divergence or at a global minimum for the residuals is the only resolution to achieve an “accurate” result. As can be seen in FIGURE 34, increasing the mesh size to roughly 1 million cells do not improve the pressure distribution either as there is a significant margin of error to the original data. After superimposing C_p^* on the original data, the finer mesh simulation indicates a much stronger shock which forms at $\sim 24\%$ chord aft of the leading edge whereas the original data suggests

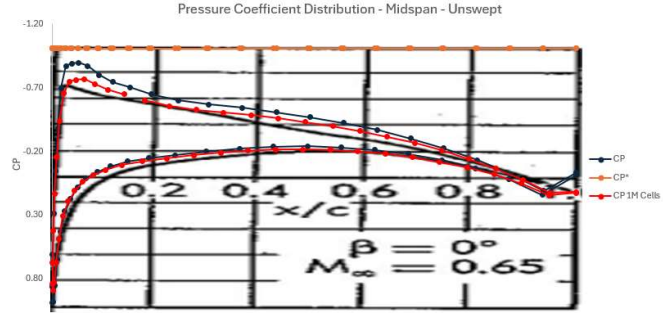


FIGURE 32 – C_p Distribution – Midspan. $\phi = 0^\circ$; $t/c=12.0\%$ $M_\infty = 0.65$.

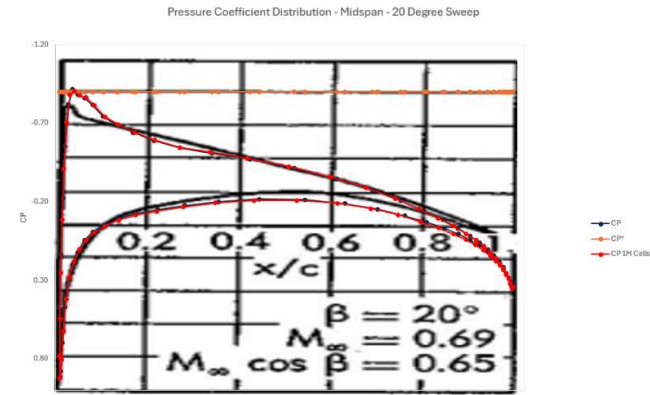


FIGURE 33 – C_p Distribution – Midspan. $\phi = 20^\circ$; $t/c=11.3\%$; $M_\infty = 0.69$.

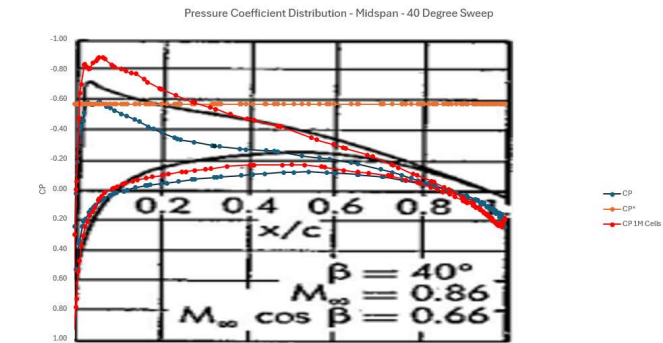


FIGURE 34 – C_p Distribution – Midspan. $\phi = 40^\circ$; $t/c=9.2\%$; $M_\infty = 0.86$.

a much weaker shock that generates at $\sim 20\%$ chord aft of the leading edge. This contrasts with the coarse mesh which indicates that the wing is just barely supercritical at the leading edge.

Considering that the original data was produced in a wind tunnel, viscous effects would provide a level of damping which will deflate the pressure distributions as well. We claim that the pressure distributions presented in FIGURES 32-34 confirm Busemann's results.

Critical flow qualities being the strong three-dimensionality demonstrated by the non-uniform isobar pattern and the formation and prominence of the shock are captured accurately. These characteristics will be retained and improved upon in a higher resolution model.

VI. Problems with taking the Center-Line C_p data as "Quasi 2-D"

Küchemann's C_p^* equation (see Eq. 5) can predict the presence of supercritical (shock prone) flow. Increasing α tends to accelerate incipient shock formation because it increases the magnitude of suction near the leading edge.

A. Practical Limitations of Küchemann's Quasi-2-D swept C_p^* equation

Let us next consider $\alpha = 5^\circ$ solutions. In FIGURE 35, with $M_\infty=0.77$ and $\varphi = 0^\circ$, $C_{p_{min}} \sim -0.63$ is found broadly around the mid chord. Since $C_p^* = -0.525$, we realize that this flow is supercritical and features a shock wave spanning the whole upper surface of the wing at the mid-chord.

In FIGURE 36, with $M_\infty=0.82$ and $\varphi = 20^\circ$, $C_{p_{min}} \sim -0.67$ is found near the downwind wall. On the centerline, $C_{p_{min}} \sim -0.5$. Since $C_p^* = -0.46$, we realize that this flow is supercritical and features a shock wave spanning the whole wing; but the shock gets much stronger near the downwind wall.

In FIGURE 37, with $M_\infty=0.5$ and $\varphi = 0^\circ$, $C_{p_{min}} \sim -2.2$ is found near the leading edge. Since $C_p^* = -2.2$, we realize that this flow is just reaching criticality along the leading edge.

In FIGURE 38, with $M_\infty=0.532$ and $\varphi = 20^\circ$, overall $C_{p_{min}} \sim -4.2$ and is found near the downwind leading edge. On the centerline, $C_{p_{min}} \sim -2.0$. Since $C_p^* = -1.88$, we realize that this flow is supercritical and features a shock wave spanning the whole wing; but the shock gets much stronger near the leading edge at the downwind wall.

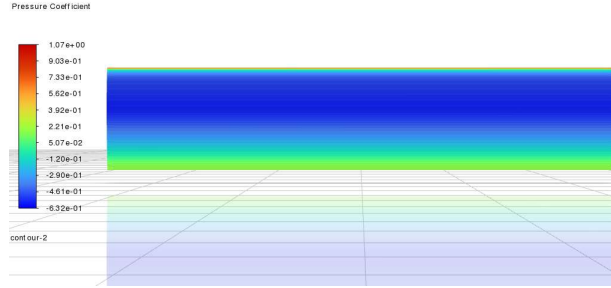


FIGURE 35 – 3-D C_p Contours. $\varphi = 0^\circ$; $t/c=12.0\%$; $\alpha = 0^\circ$. $M_\infty = 0.77$.

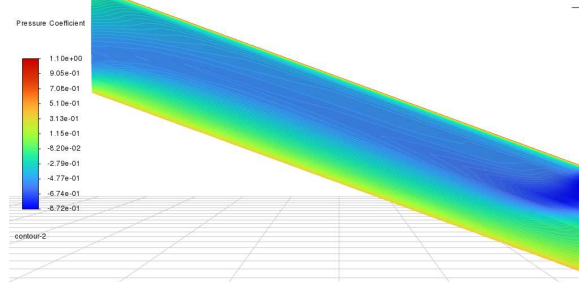


FIGURE 36 – 3-D C_p Contours. $\varphi = 20^\circ$; $t/c=11.3\%$; $\alpha = 0^\circ$. $M_\infty = 0.82$.

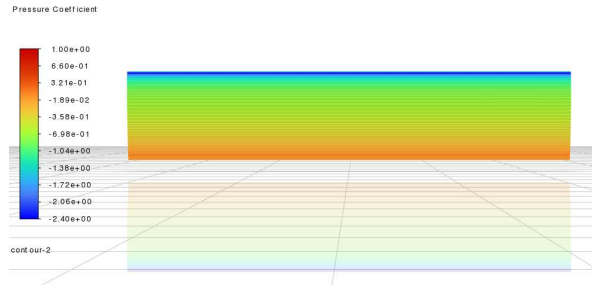


FIGURE 37 – 3-D Mach Contours. $\varphi = 0^\circ$; $t/c=12.0\%$. $\alpha = 5^\circ$. $M_\infty = 0.5$.

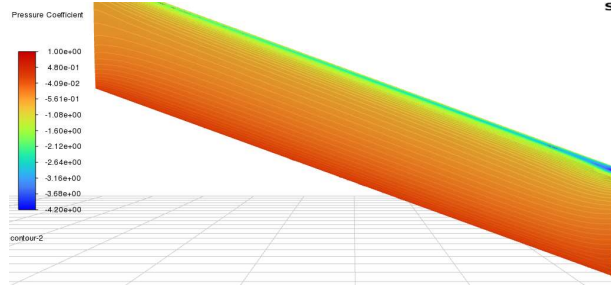


FIGURE 38 – 3-D C_p Contours. $\varphi = 20^\circ$; $t/c=11.3\%$ $\alpha = 5^\circ$. $M_\infty = 0.532$.

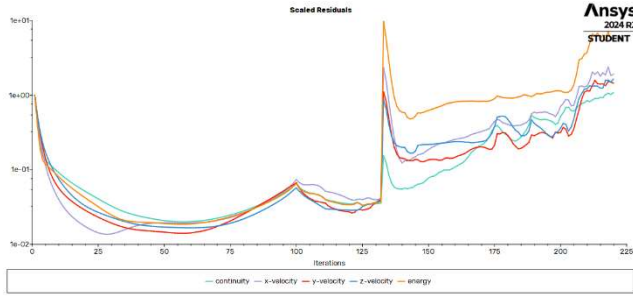


FIGURE 39 – “Convergence” history of a divergent solution. $\phi = 40^\circ$; $t/c = 9.2\%$. $\alpha = 5^\circ$. $M_\infty = 0.9$

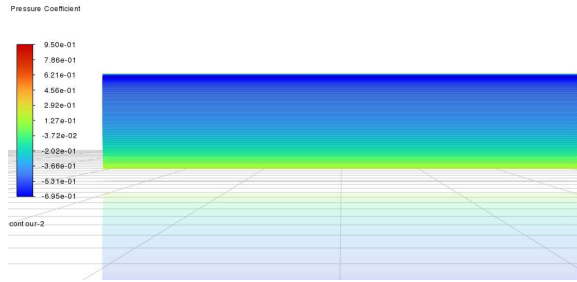


FIGURE 40 – 3-D C_p Contours. $\phi = 0^\circ$; $t/c = 12.0\%$. $\alpha = 2^\circ$. $M_\infty = 0.2$

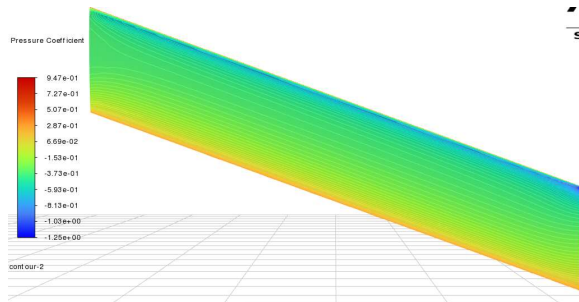


FIGURE 41 – 3-D C_p Contours. $\phi = 20^\circ$; $t/c = 11.3\%$; $\alpha = 2^\circ$. $M_\infty = 0.212$.

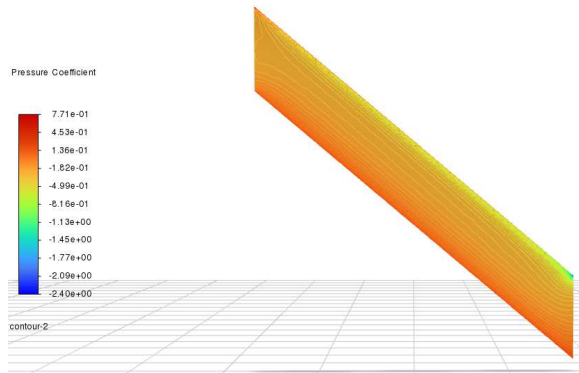


FIGURE 42 – 3-D C_p Contours. $\phi = 40^\circ$; $t/c = 9.6\%$. $\alpha = 2^\circ$. $M_\infty = 0.261$

At higher sweeps and speeds, we were unable to reliably converge solutions to attain criticality at the midspan without “virtual tunnel choking.” As the whole control-volume chokes, it creates such disturbed flow conditions that the CFD solution tends to diverge; see FIGURE 39.

B. Is the Flow field Three-Dimensionality a Transonic Effect?

Next, we show how general are the Quasi-2-D assumptions at low speeds. Generally, across all sweep angles showcased in FIGURES 40-42, the isobars look straight. There is some variation seen in the swept cases which is due to the proximity of the “virtual wind tunnel walls.” However, for a majority of the span, the isobars are straight and follow the sweep angle. This begs the question as to whether incompressible flow conditions permit a quasi-2-D assumption. In addition, is three-dimensionality simply relegated to a non-linear isobar pattern?

C. Is Three-Dimensionality a curved isobar issue, or something else yet?

Despite the iso-bar lines appearing relatively straight, 2-D does not accurately reflect the 3-D characteristics experienced. Ira H. Abbott, a notable aerospace engineer who worked at the NACA Langley Memorial Aeronautical Laboratory (LMAL) along with others such as Albert von Doenhoff, would go on to say that “Wing design depends on many three-dimensional effects, not just an airfoil section.”[39]

FIGURES 43 and 44 (overleaf) compare the pure 2-D pressure distribution to a 2-D slice taken from the 3-D solution along the midspan. Here, we see a significant difference between a 2-D CFD simulation and the 3-D model. The 2-D model greatly overestimates the pressure distribution and when in the perspective of aircraft design, would show that the wing produces somewhat more lift than it does. For low-speed applications, this error may not be severe since C_p^* is generally very high. For an unswept wing at a freestream Mach of 0.2, $C_p^* = -16.31$, making it extremely unlikely for wings to every reach criticality in this condition.

The assumption of quasi-2-D flow in high subsonic regimes, particularly as perturbations begin to form, can result in considerable errors. FIGURE 44 demonstrates how the assumption that 2-D flows represent real flows fails. A designer following the 2-D Critical Mach Number predictions will develop underperforming wings. In this particular case, with an unswept section the 2-D case has a **lower Critical Mach Number** than the “virtual wind tunnel” spanning 3-D wing coupon.

FIGURE 44 shows that the 2-D airfoil run at $M=0.65$ is slightly supercritical (it has a Critical Mach Number, $M_{cr} \sim 0.63$ as its peak suction has $C_{p_{min}} \sim -1.1$ but $C_p^* = -1.09$). The “2-D slice” out of the midspan of the 3-D wing has a lower peak suction $C_{p_{min}} \sim -0.7$ which implies a wing with $M_{cr} \sim 0.72$. This 0.09 difference in Critical Mach Number is substantial and highlights the sensitivity of pressure distributions and lift predictions to compressibility effects in high subsonic flows, emphasizing the limitations of 2-D approximations in these cases.

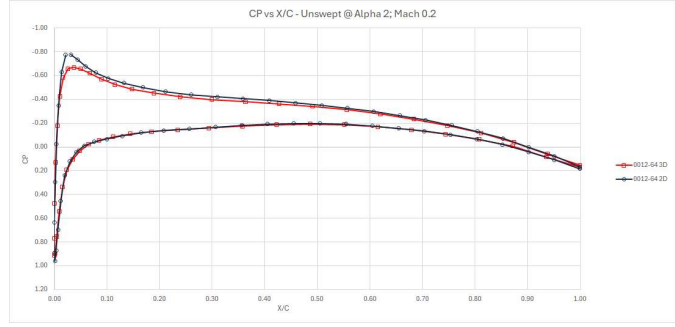


FIGURE 43 – 2-D vs 3-D Pressure Coefficient Distribution for a 0° Swept Wing (2-D vs midspan 3-D, Mach 0.2, $\alpha = 2^\circ$)

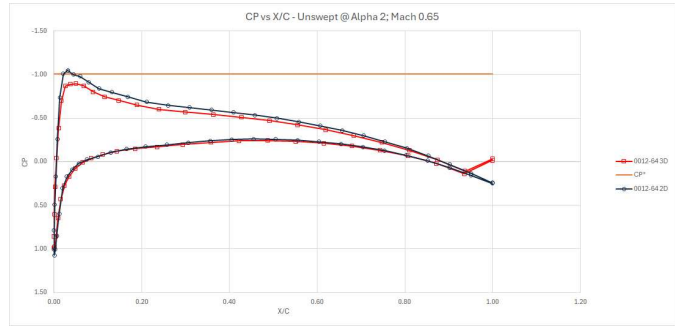


FIGURE 44 – 2-D vs 3-D Pressure Coefficient Distribution for a 0° Swept Wing (2-D vs midspan 3-D, Mach 0.65, $\alpha = 2^\circ$)

VII. Conclusion

We believe that the NACA 0012-64 airfoil serves as an adequate approximation of the original R4009 used by Busemann, and Lippisch & Beuschausen.[1][2][21][44]

Analyzing the three-dimensionality in the pressure distributions developed in our CFD based “virtual wind tunnel” reveals that Busemann’s Simple Sweep Theory provides only an approximate description of reality. This theory suggests that the aerodynamic characteristics of a swept wing can be derived from its 2-D airfoil section; however, it does not account for the complex interactions that occur in three-dimensional flow fields. As such, we assert that while Simple Sweep Theory has been championed by notable figures such as Jones [26][27][28][29] and Anderson [5][6], it is inherently imprecise and can lead to oversimplifications in wing design analysis.

Despite its limitations, the prominence of Anderson’s texts [5][6] continues to have an enduring popularity in academia and industry. His explanations and insights into aerodynamic principles have made Simple Sweep Theory a foundational concept in many aerodynamic studies, albeit sometimes at the cost of considering more complex phenomena.

Küchemann’s work encapsulates the three-dimensionality that we see in real-world applications and offers a more nuanced understanding of aerodynamic behavior. With this in mind and given the prevalence of Anderson’s works, Küchemann serves as a proper graduation in providing a better understanding of aircraft design.

While 2-D and quasi-2-D data provides valuable insight for initial configurations, we see that the omission of three-dimensional effects when selecting wing sections can easily lead to a design astray. Over reliance on 2-D and quasi-2-D results will particularly fail when applied to intricate planforms. The emergence of unexpected 3-D pressure

anomalies the create unintended shock waves can also lead to premature low separation. These issues will manifest themselves in terms of shortfalls in aircraft system performance in terms of high drag or reduced stability and controllability.

The importance of understanding three-dimensional flow is amplified in high-performance aircraft or more complex shapes such as serrated wings, where nuanced geometry leads to major changes in aerodynamic behavior. Getting these trends right at the outset makes the difference between success and failure.

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